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SUPERSEDING
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April 7, 1944

FEDERAL SPECIFICATION

PAPER, FILTER, ANALYTICAL

This specification was approved on the above date by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope.—This specification covers only the analytical filter papers most commonly used in the laboratories of the Federal Government and does not cover those used in relatively small amounts such as fluted papers.

1.2 Types and classes.—Filter paper covered by this specification shall be of the following types and classes, as specified:

Type I.—Qualitative.

Class 1.—For coarse and gelatinous precipitates.

Class 2.—For medium precipitates.

Class 3.—For fine precipitates.

Class 4.—For fine precipitates, hardened.

Type II.—Quantitative.

Class 5.—For coarse and gelatinous precipitates.

Class 6.—For medium precipitates.

Class 7.—For fine precipitates.

Class 8.—For fine precipitates, thin.

1.3 Shape and sizes.—Unless otherwise specified, the filter paper shall be furnished in circles of the following diameters, as specified:

5.5, 7, 9, 11, 12.5, 15, or 18.5 centimeters.

A tolerance of plus or minus 0.2 centimeters in size will be permitted.

2. APPLICABLE SPECIFICATIONS

2.1 The folowing Federal Specifications, of the issues in effect on date of invitation for bids, form a part of this specification:

NN-B-601—Boxes; Wood-Cleated-Plywood, for Domestic Shipment.

NN-B-621—Boxes; Wood, Nailed and Lock-Corner.

NN-B-631—Boxes; Wood, Wirebound, for Domestic Shipment.

UU-P-31 —Paper; General Specifications and Methods of Testing.

2.2 Specifications and other publications applicable only to individual departments are listed in section 7.

3. REQUIREMENTS

3.1 Material.—The filter paper shall be composed of 100-percent fully bleached fibers.

3.2 Apha-cellulose. — The alpha-cellulose content shall be not less than 95 percent.

3.3 Copper number.—The copper number shall be not more than 0.5.

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3.4 Acidity.—The pH value shall be not less than 6.0 nor more than 8.0.

3.5 Ash.—The ash of the circles of filter paper of all classes shall not exceed the amounts specified in table I. The ash of other sizes shall be of the same proportion.

TABLE I.—*Maximum ash content of circles*

Diameter of circle	Ash		
	Type I All classes	Type II, Classes 5, 6, and 7	Type II, Class 8
Centimeters	*Milligrams*	*Milligrams*	*Milligrams*
5.5	0.25	0.025	0.019
7.0	0.41	0.041	0.031
9.0	0.67	0.067	0.051
11.0	1.00	0.100	0.076
12.5	1.29	0.129	0.098
15.0	1.86	0.186	0.141
18.5	2.83	0.283	0.215

3.6 Bursting strength, time of flow of water, and retention of precipitates.—The bursting strength, time of flow of water, and retention of precipitates shall comply with the requirements of table II.

TABLE II.—*Average bursting strength, average time of flow of water, and retention of precipitates*

Class	Bursting strength, wet, minimum	Time of flow of water, maximum	Retention of precipitates
	Points	*Seconds*	*Filtrate clear from*
1	3	20	Ferric hydroxide.
2	3.5	40	Lead sulfate.
3	4	150	Barium sulfate.
4	60	260	Barium sulfate.
5	3	20	Ferric hydroxide.
6	3.5	40	Lead sulfate.
7	4	150	Barium sulfate.
8	4	150	Barium sulfate.

3.7 Type I, class 4.—Type I, class 4, filter paper shall have a surface sufficiently hard to permit precipitates to be scraped off.

3.8 Workmanship.—The filter paper circles shall have uniform formation and shall be clean and free from imperfections which may affect their serviceability.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Sampling.—Samples shall consist of five cartons, each taken from a different shipping container of a delivery of five or more containers, or one carton taken from each of any lesser number of containers in a delivery.

4.2 Inspection.—Inspection of samples shall be made to determine compliance with this specification with respect to type, class, size, workmanship, packaging, packing, and marking. (See 1.2, 1.3, 3.8, 5.1, 5.2, and 5.3.)

4.3 Tests.—Specimens from each carton of the sample shall be tested.

4.3.1 The following tests shall be made in accordance with the methods described in Federal Specification UU–P–31:

Tests	Method No.
Fiber identification and quantitative determination	300
Alpha-cellulose	201
Copper number	220
Acidity, pH	200

4.3.2 *Ash.*—The necessity for a very accurate determination of ash content, together with the small amount present, requires special modifications of the standard method for testing paper. A large test specimen but as small an ignition crucible as possible are required for good accuracy. These conditions are fulfilled by wetting the filter paper with distilled water, thus permitting compact wadding of the paper for conservation of space. The use of a tare crucible and special precautions in the testing operations are necessary. The test shall be made as follows:

Use 20-milliliter platinum crucibles with covers. A tare crucible with a cover is carried

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through all operations exactly the same as the crucibles used for the tests. The tare crucible must have practically the same weight and surface area as the test crucible.

Heat the crucibles and covers in a muffle furnace at approximately 925°C. Cool in a desiccator and weigh to the nearest 0.1 milligram. Dry about 6 grams of the paper to constant weight at 100° to 105°C., and weigh to the nearest 0.01 gram. Wet the paper with distilled water containing not more than 10 parts per million of igneous solids, then roll it into a compact mass with clean fingers. Put the test specimen in a crucible, cover it, and place it in a cold muffle furnace. Start the ignition and after smoking has ceased remove the cover and continue the ignition for 2 hours at 925°C. Replace the cover, cool, and weigh the crucible and ash. The ignition is considered to be complete when the weight of the ash becomes constant after reignition for 30 minutes. All the cooling must be done in a desiccator, and the test crucibles and tare crucible must be allowed to remain on the balance pans for the same length of time. The ash must be protected from air currents at all times.

Duplicate determinations shall be made, and they shall agree within 0.003 percent of the weight of the dry paper. Report the average of the values to the nearest 0.01 milligram per 11-centimeter circle of the paper.

4.3.3 *Bursting strength, wet.* — Clip together a pack of five circles of the filter paper, and immerse the pack in distilled water at 23°C. for 5 minutes. Remove the pack, allow to drain suspended for 10 seconds, then immediately determine the bursting strength in accordance with method 112 described in Federal Specification UU–P–31. Use enough packs to obtain not less than 10 values. Report the bursting strength to the nearest ¼ point.

4.3.4 *Time of flow of water.*—Prepare distilled water by prefiltering it through filter paper that filters at least as slowly as the paper to be tested. Use two or more filters in series. For each kind of filter paper to be tested, a volume of prefiltered water equal to three or four times the volume of the cone will usually be sufficient. If a considerable quantity of water is to be prepared, the filters should be renewed frequently, because they lose their effectiveness with continued flow of water through them.

Carefully fold a filter circle in the usual way to form a 60° cone. Place it in a 60° funnel and fill it with water prepared in the manner described above. Press down the folds to expel all air pockets and to make the three plies smooth and in good contact. Allow about three-fourths of the water to filter through, then pour off the excess. Save the water to wet the next specimen. Remove the empty cone and suspend it freely over a burette by supporting the cone in a loop of wire of such size that it will support the cone at about two-thirds of the distance from its apex.

Using prefiltered water at a temperature of 23° ± 2°C., pour into the cone, all at once, a measured volume which is approximately two-thirds of the volume of the cone. When one-fifth of the water has filtered through into the burette, start a stopwatch. When half the remaining volume has filtered through, stop the stopwatch and record the time in seconds. Test at least 10 circles, using the same water over and over and adding to it from the prepared supply as found necessary.

The time required to filter half the volume of water remaining in the conical filter is substantially the same for any given paper, irrespective of the size of the test specimen. Suitable volumes of water initially poured in for filter paper circles of the sizes most generally used are as follows:

Diameter of circle	*Initial volume*
Centimeters	*Milliliters*
11	25
9	15
7	7.5
5.5	3.5

Report the filtration time in seconds and specify the diameter in centimeters of the filter paper used for the test.

Note 1.—For some reason that has never

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been explained satisfactorily, the rate of flow of liquids through filter paper, and other porous materials, becomes progressively slower with continued flow. It has been found, however, that the change in rate can be almost eliminated for filter paper by prefiltration of the water used in making the test.

4.3.5 *Retention of precipitates.*—Retention of precipitates shall be determined by examining the filtrate, which shall be caught in an Erlenmeyer flask, for any precipitate that may pass through the filter paper. Swirl the filtrate to bring any precipitate to the center and view from above against a black background. The precipitates shall be prepared as follows:

4.3.5.1 *Ferric hydroxide, cold.* — Add a slight excess of dilute ammonium hydroxide to cold ferric chloride solution and shake. Make fresh each time and use immediately.

4.3.5.2 *Lead sulfate, cold.*—Prepare a solution of lead acetate by dissolving 10 grams of the anhydrous salt or 11.5 grams of the trihydrate in water and dilute to 100 milliliters in a graduated cylinder. Filter through a class 7 filter paper and add 40 milliliters of 6*N* sulfuric acid and 80 milliliters of alcohol. Filter after standing for 4 hours. Wash the precipitate with dilute sulfuric acid (5 milliliters concentrated acid in 100 milliliters of water), then with alcohol.

4.3.5.3 *Barium sulfate, hot.*—Dissolve 0.55 gram of K_2SO_4 in 275 milliliters of water and add 1.0 milliliter of concentrated HCl. This solution contains the equivalent of about 0.74 gram of $BaSO_4$. Heat to boiling, and while at the boiling temperature, add slowly and with continuous stirring 25 milliliters of a 5-percent solution of $BaCl_2.2H_2O$. It is very important to have the specified concentration of acid and to keep the solution at the boiling temperature during the precipitation. Let stand without agitation in a boiling water bath for 2 hours ± 15 minutes. Stir the precipitate until it is evenly suspended in the liquid, and filter approximately 50 milliliters of the mixture through each circle of paper.

4.3.6 *Diameter of circles.*—The diameter of not less than 10 circles shall be measured and the average of the measurements reported to the nearest 0.1 centimeter.

5. PREPARATION FOR DELIVERY

5.1 Packaging.—Circles of filter paper shall be packaged in cartons, 100 per carton. Other forms of filter paper shall be packaged according to commercial practice.

5.2 Packing. — Packages of filter paper shall be overpacked in an exterior container constructed in accordance with the requirements of Federal Specification NN–B–601, NN–B–621, or NN–B–631. Unless otherwise specified, gross weight of the container shall not exceed approximately 100 pounds. Unless otherwise specified, only one type, class, and size of filter paper shall be packed in each shipping container.

5.3 Marking.

5.3.1 *Cartons.* — Each package shall be marked with the stock number, name of the commodity, the type, class, unit, and quantity contained therein, and the name of the contractor and manufacturer. In addition, each carton containing class 5, class 6, class 7, and class 8 filter papers shall be marked with the weight of ash per circle.

5.3.2 *Shipping containers.*—Shipping containers shall be marked on one side with the name of the commodity, the type, class, size, and quantity contained therein, as defined by the contract or order under which the shipment is made, the name of the contractor, and the number of the contract or order.

6. NOTES

6.1 Ordering data.—Purchasers should exercise any desired options offered herein and should specify the type, class, and size of filter paper required. (See 1.2, 1.3, and 7.2.5.)

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6.2 Bid samples.—If samples are required with bid, this requirement should be stated in the inquiry or invitation for bids, giving information as to the size or quantity of sample.

6.3 Requirements for alpha-cellulose content, copper number, and acidity are included in this specification because they are of importance relative to the purity of the cellulose and the stability of the papers. A high alpha-cellulose content and a lower copper number are characteristic of cellulose of a high degree of purity, that is, cellulose that has not been degraded appreciably. Degraded cellulose is partially soluble in some solutions and therefore may pass into filtrates and cause erroneous analytical results. Both degraded cellulose and high acidity result in unstable paper that may become too brittle for satisfactory use. For additional information on this and on the special test methods described, the following publications, respectively, of the National Bureau of Standards should be consulted:

Research Paper RP1809, Development of Standards for Analytical Filter Papers.
Research Paper RP1653, Methods for the Evaluation of Analytical Filter Papers.

Patent notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

7. DEPARTMENTAL REQUIREMENTS

7.1 The following specifications and other publications, of the issues in effect on date of invitation for bids, and special requirements, form a part of this specification for purchases made under this specification by the respective departments.

7.2 Army, Navy, and Air Force.

7.2.1 *Applicable specifications and other publications.*

Military Specifications:

JAN–P–105—Packaging and Packing for Overseas Shipment — Boxes, Wood, Cleated, Plywood.

JAN–P–106—Packaging and Packing for Overseas Shipment — Boxes; Wood, Nailed.

JAN–B–107—Boxes, Wood, Wirebound (Overseas Type).

JAN–P–125—Packaging and Packing for Overseas Shipment — Barrier-Materials, Waterproof, Flexible.

JAN–P–140—Packaging and Packing for Overseas Shipment — Adhesives, Water-Resistant, Case-Liner.

U. S. Army Specifications:

94–40645—Marking, Exterior, Domestic and Export Shipment, by Contractors.[1]

100–2 —Standard Specification for Marking Shipments by Contractors.[2]

Navy Department Specification:

General Specifications for Inspection of Material.[3]

Bureau of Supplies and Accounts Publication:

Navy Shipment Marking Handbook.[3]

(Copies of specifications required for Army purchases may be obtained upon application to the procuring agency. Both the title and identifying symbol or number should be stipulated when requesting copies.

(Copies of Federal and Military specifications required for Navy purchases, Navy Department specifications, and the Navy Shipment Marking Handbook

[1] Applicable to Air Force purchases only.
[2] Applicable to Army purchases only.
[3] Applicable to Navy purchases only.

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may be obtained without cost upon application to the Bureau of Supplies and Accounts, Navy Department, Washington 25, D. C., except that Naval activities should make application to the Supply Officer in Command, Naval Supply Center, Norfolk 11, Va. Both the title and identifying symbol or number should be stipulated when requesting copies of specifications.

(Copies of specifications required for Air Force purchases may be obtained upon application to the Commanding General, Air Matériel Command, Wright-Patterson Air Force Base, Dayton, Ohio. Both the title and identifying symbol or number should be stipulated when requesting copies.)

7.2.2 *Packaging.*—One hundred sheets of filter paper constitute one unit. Each unit shall be packaged in accordance with the requirements of 5.1. A number of unit packages as specified in the purchase order, contract, or invitation for bids, shall be further packaged in a carton of appropriate size and design, or be completely wrapped with kraft paper of suitable weight. Carton or wrapping shall be adequately secured to prevent accidental opening.

7.2.3 *Packing for overseas shipment.*—A number of intermediate packages or unit cartons as specified in the contract, purchase order, or invitation for bids, shall be overpacked in an exterior container designed for type 2 load, and constructed in accordance with the requirements of Military Specification JAN–P–105, JAN–P–106, or JAN–B–107. Closure and strapping shall comply with the appendix to the applicable specification. Each box shall be lined with a sealed waterproof barrier (bag) fabricated of material conforming to requirements of Military Specification JAN–P–125. Closure shall be completely sealed with a continuous seam, at least ¾ inch wide, of water-resistant adhesive complying with Military Specification JAN–P–140. Unless otherwise specified, gross weight of the container shall not exceed 100 pounds.

7.2.3.1 *Packing variation permitted.*—If the number of units in any shipment is not the exact multiple of the number of units specified to be contained in each shipping container, the units not constituting the nearest such multiple shall be packed in an exterior container of suitable size and design and so constructed as to insure safe delivery to specified destination.

7.2.4 *Marking.*—In addition to the marking specified in 5.3, shipping containers for the Army shall be marked in accordance with the requirements of U. S. Army Specification 100–2; for the Air Force, in accordance with U. S. Army Specification 94–40645; and for the Navy, in accordance with the Navy Shipment Marking Handbook.

7.2.5 *Ordering data.* — Requests, requisitions, schedules, and contracts or orders should specify whether the subject commodity is to be packaged, packed, and shipping containers marked for overseas shipment, and whether shipping containers are to be marked for Army, or Navy, or Air Force.

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(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)